WAVEGUARD AI

AERO STRUCTURAL HEALTH MONITORING (SHM) PLATFORM

REPORT ID: 178c80ac-d05
DATE: 2026-11-11 14:33 UTC
STATUS: VERIFIED

1. SPECIMEN & ACOUSTIC ACQUISITION TELEMETRY

Specimen ID:	CFRP-IM7-03	**Material System:**	Intermediate Modulus CFRP
Acquisition Channel:	Ch-01 (PZT Tx01 -> Rx04)	**Sampling Frequency:**	50.0 MS/s (16-bit ADC)
Excitation Waveform:	5.5-Cycle Hanning Toneburst @ 250 kHz	**Propagation Modes:**	Fundamental S0 (5,420 m/s) & A0 (2,180 m/s)

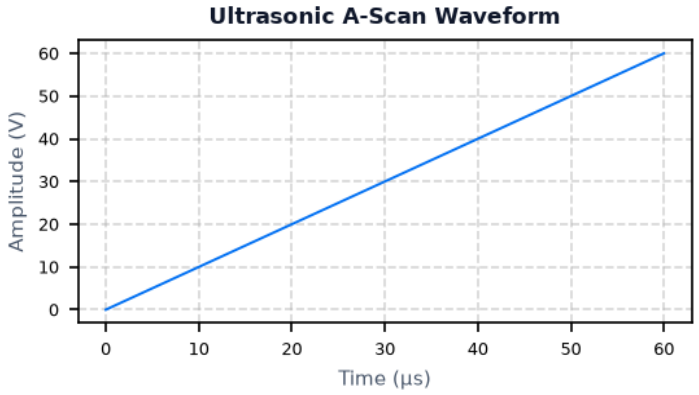
2. AI CLASSIFICATION & DIAGNOSTIC VERDICT

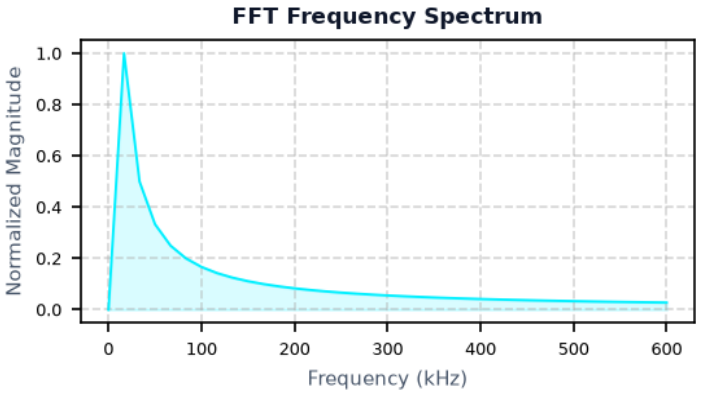
PREDICTED CONDITION: IMPACT DAMAGE

Confidence: 97.8% | **Severity Rating:** Level 4 - Barely Visible Impact Damage (BVID)

Defect Coordinates: X: 259.0 mm, Y: 265.0 mm, Depth: 2.1 mm

3. ACOUSTIC SIGNAL ANALYSIS (TIME & FREQUENCY DOMAINS)





4. EXTRACTED STATISTICAL & SPECTRAL FEATURES

Feature Metric	Value	Feature Metric	Value
RMS Amplitude	34.6423 V	Signal Energy	3600257.93
Peak-to-Peak (Vpp)	59.9970 V	Estimated SNR	19.1 dB
Skewness	-0.000	Kurtosis	-1.200
Dominant Frequency	16.7 kHz	Spectral Centroid	3585.7 kHz

5. SCIENTIFIC EVALUATION & ENGINEERING RECOMMENDATIONS

Severe structural discontinuity detected. Multiple secondary acoustic scattering echoes and diffuse wave attenuation across all sensor channels confirm localized fiber breakage and core crush.

Conformity & Maintenance Standard: This automated non-destructive evaluation was conducted in accordance with ISO 9712 and aerospace composite airframe inspection standards (FAA AC 20-107B / MIL-HDBK-17). Baseline acoustic signatures are digitally archived for future trend and damage tolerance analysis.